Rambus

Security IP
EIP-130

Root of Trust engine
Driver Development Kit v3.1

Test Report

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1 Summary

This document describes the test results for the *EIP-130* DDK.

1.1 Related documents

The following documents are part of the documentation set for the *EIP-130 Root of Trust engine* Driver Development Kit.

Ref.	Title	Document Number
[1]	EIP-130 DDK Test Specification	007-130310-310
[2]	README Regression System	N.A.
[3]	EIP-130 Hardware Reference Manual	007-130xx0-201[1]
[4]	EIP-130 Firmware Reference Manual	007-130xx0-204[1]
[5]	EIP-130 Driver User Guide	007-130310-301
[6]	EIP-130 DDK User Guide	007-130310-308

[1] xx refers to the product version's major/minor version digits. So, for example version 3.0 translates to 30

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1.2 Target Audience

This document is intended for the driver developers and users. This document assists the driver developer in testing the DDK components for the *Root of Trust engine*.

2 Verification Environment

The following HW and SW platforms and toolkits were used for the DDK verification:

- ARM Aarch32 - Xilinx® Zynq®-7000 SoC ZC706 evaluation board
  - Dual ARM® Cortex®-A9 core,
  - 32-bit Linux Kernel (linux-4.4-zynq-zc706),
  - GNU C compiler version 5.2.1 for ARM.
- x86-64
  - 8 processors: Intel® Xeon® CPU E5507, clock frequency 2.27GHz,
  - Linux 5.4.0 (non-)debug 64-bit preemptive SMP configuration kernel,
  - GNU C compiler version 9.3.0,
  - Xilinx® Virtex6 FPGA PCIe board 50MHz
- ARMv8 Aarch64 - Juno ARM® Development Platform
  - ARM® Cortex®-A57 and Cortex®-A53 MPCore for ARMv8 big.LITTLE™,
  - 64-bit Linux Kernel with big.LITTLE support,
  - GNU C compiler version 4.8.3 for ARM,
  - LogicTile Express 20MG - single Virtex-7 FPGA 75MHz.

The DDK verification is mainly done with following HW/FW combination:

- *EIP-130* HW3.0.15 and *EIP-130* FW3.2 (used as base for the numbers in section 5)
- *EIP-130* HW2.3.2 and *EIP-130* FW2.5.19

Note that older firmware versions were also used to check the backwards compatibility.

The following sections describe the performed test cases that were executed on the FPGA platforms. For the FPGA platform, the tests were executed using both the DDK driver kernel-space and user-space driver deployment.

3 Pre-defined Configurations

For the ease of use of this document, Table 1 shows the pre-defined configurations as mentioned in the *EIP-130 DDK Test Specification* [1] for the Driver, Demo Applications and Test Tool.

Table 1 Pre-defined driver and test tool configurations

Driver Configuration[1]	Polling / Interrupts	Bounce Buffers	Performance
C0	Polling	Off	Yes
C1	Interrupts	Off	Yes
C2	Polling	Off	No
C3	Polling	On	No
C4	Interrupts	On	Yes
C5	Interrupts	On	No

[1] Endianness Conversion and TokenID used to check DMA ready applies to all Driver Configurations when enabled.

4 Test Results

This chapter describes the results of all the performed tests on all the HW platforms used in the verification environment. A test case can either pass or fail. The following indication is used.

P = Passed test case

F = Failed test case

These tests were performed using Linux debug and non-debug kernel configurations. Refer to [2] for details on the used debug configuration parameters for the Linux kernel.

Test cases were executed in the following deployments. Pass means that the test has passed in all deployments for which the test was carried out:

- Driver and test application both in kernel space. These tests were carried out for all driver configurations.
- Driver and test application both in user space. These tests were carried out for all driver configurations.

4.1 VAL Test Tool Results

The next sections describe the results of the related test cases for the *EIP-130 Root of Trust engine* also-called *VaultIP* or *EIP-130*. The test results are obtained from the test logs created during the regression tests running on the given hardware platform. Refer to [2] for details on how to execute the demo applications and test tools.

4.1.1 Xilinx® Zynq®-7000 SoC ZC706 Evaluation Board Results

Table 2 below lists the results of the VAL Test Tool that was executed on a Xilinx® Zynq®-7000 SoC ZC706 evaluation board. The VAL Test Tool runs the test suite, test cases and tests specified in [1] for the pre-defined configurations.

Table 2 Xilinx® Zynq®-7000 SoC ZC706, VAL Test Tool results

Driver Configuration[1]	Polling / Interrupts	Bounce Buffers	Performance	Result (P/F)
C0	Polling	Off	Yes	P
C1	Interrupts	Off	Yes	P
C2	Polling	Off	No	P
C3	Polling	On	No	P
C4	Interrupts	On	Yes	P
C5	Interrupts	On	No	P

[1] Endianness Conversion and TokenID used to check DMA ready applies to all Driver Configurations when enabled.

4.1.2 x86-64 SMP system with Xilinx® Virtex6 FPGA Results

This DDK release is built for the Xilinx® Virtex6 FPGA environment only. The VAL Test Tool is not executed on this specific board.

4.1.3 Juno ARM® Development Platform with LogicTile Express Results

This DDK release is built for the Juno ARM® Development Platform environment only. The VAL Test Tool is not executed on this specific board.

4.2 Demo Application Results

The following are the results of the tests for the Demo Application `da_val_*`.

4.2.1 Xilinx® Zynq®-7000 SoC ZC706 Evaluation Board Results

Table 3 Xilinx® Zynq®-7000 SoC ZC706, Demo Application results

Driver Configuration[1]	Polling / Interrupts	Bounce Buffers	Performance	Result (P/F)
C0	Polling	Off	Yes	P
C1	Interrupts	Off	Yes	P
C2	Polling	Off	No	P
C3	Polling	On	No	P
C4	Interrupts	On	Yes	P
C5	Interrupts	On	No	P

[1] Endianness Conversion and TokenID used to check DMA ready applies to all Driver Configurations when enabled.

Pass means that the demo application `da_val_*` passed all tests.

4.2.2 x86-64 SMP system with Xilinx® Virtex6 FPGA Results

This DDK release is built for the Xilinx® Virtex6 FPGA environment only. The Demo Application is not executed on this specific board.

4.2.3 Juno ARM® Development Platform with LogicTile Express Results

This DDK release is built for the Juno ARM® Development Platform environment only. The Demo Application is not executed on this specific board.

4.3 Memory Leak Tests

The driver has its own memory administration implemented in the Driver Framework and unloading the driver triggers inspection of this administration for un-freed resources. There are no memory leaks when the driver does not report them in the test logs. The test logs are inspected automatically by the regression system for the memory leak reports.

Note: The failed secure timer tests were expected due to their time limitation.

4.3.1 Xilinx® Zynq®-7000 SoC ZC706 Evaluation Board Results

No memory leaks reported by the driver for all tests.

4.3.2 x86-64 SMP system with Xilinx® Virtex6 FPGA Results

This DDK release is built for the Xilinx® Virtex6 FPGA environment only.

4.3.3 Juno ARM® Development Platform with LogicTile Express Results

This DDK release is built for the Juno ARM® Development Platform environment only.

5 Code Characteristics

5.1 Code size

5.1.1 Xilinx® Zynq®-7000 SoC ZC706 Evaluation Board

The modules in the table below are cross compiled for the ZC706 Evaluation Board Linux Aarch32 non-debug SMP kernel and DDK configuration C0.

Table 4 DDK modules footprint data for Aarch32 (bytes)

Module	Deployment	Text	Data	BSS	Total
Driver Combined `driver_vip_k.ko`	Kernel-space	104087	616	1856	106559
Global Control Driver `driver_gc_k.ko`	Kernel-space	57649	496	1840	59985
VAL Driver `driver_val_k.ko`	Kernel-space	104083	616	1860	106559
VAL Test Tool `test_val_*`	Kernel-space	407833	1312	3616	412761
	User-space (combined)	685746	1548	6340	693634
	User-space (proxied)	594840	1312	4948	601100
VAL Demo Application `da_val_*`	Kernel-space	4157	448	0	4605
	User-space (combined)	111228	636	1676	113540
	User-space (proxied)	20179	388	280	20847
Demo Application `da_val_specials_*`	User-space (combined)	134176	648	2716	137540
	User-space (proxied)	43488	424	1324	45236
Demo Application `da_securedebug_*`	User-space (combined)	126020	648	11812	128480
	User-space (proxied)	34971	400	416	35787
Demo Application `da_otpprogram_*`	User-space (combined)	104617	636	3780	109033
	User-space (proxied)	13753	404	2388	16545
Demo Application `da_otpkeyblob_*`	User-space (combined)	123725	636	8300	132661
	User-space (proxied)	32865	404	6908	40177

5.1.2 x86-64 SMP system with Xilinx® Virtex6 FPGA

The modules in the table below are built for Linux x86-64 non-debug SMP kernel and DDK configuration C0.

Table 5 DDK modules footprint data for x86-64 (bytes)

Module	Deployment	Text	Data	BSS	Total
Driver Combined `driver_vip_k.ko`	Kernel-space	126927	1361	3092	131380
Global Control Driver `driver_gc_k.ko`	Kernel-space	67620	1073	3056	71749
VAL Driver `driver_val_k.ko`	Kernel-space	126713	1361	3076	131150
VAL Test Tool `test_val_*`	Kernel-space	470320	2608	4736	477664
	User-space (combined)	753904	36692	8576	799172
	User-space (proxied)	652985	36260	6424	695669
VAL Demo Application `da_val_*`	Kernel-space	4310	896	0	5206
	User-space (combined)	127214	1284	2752	131250
	User-space (proxied)	26060	812	600	27472
Demo Application `da_val_specials_*`	User-space (combined)	152050	1300	3840	157190
	User-space (proxied)	51389	880	1656	53925
Demo Application `da_securedebug_*`	User-space (combined)	142598	1284	2944	146826
	User-space (proxied)	41412	828	792	43032

Module	Deployment	Text	Data	BSS	Total
Demo Application da_otpprogram_*	User-space (combined)	117687	1284	4928	123899
	User-space (proxied)	16850	844	2744	20438
Demo Application da_otpkeyblob_*	User-space (combined)	138387	1284	9536	149207
	User-space (proxied)	37582	844	7352	45778

5.1.3 Juno ARM® Development Platform with LogicTile Express

The modules in the table below are cross compiled for the ARMv8 Juno board Linux Aarch64 non-debug SMP kernel and DDK configuration C0.

Table 6 DDK modules footprint data for Aarch64 (bytes)

Module	Deployment	Text	Data	BSS	Total
Driver Combined driver_vip_k.ko	Kernel-space	103922	800	2808	107530
Global Control Driver driver_gc_k.ko	Kernel-space	56540	584	2776	59900
VAL Driver driver_val_k.ko	Kernel-space	103922	800	2808	107530
VAL Test Tool test_val_*	Kernel-space	447096	2128	4664	453888
	User-space (combined)	698106	2816	8328	709250
	User-space (proxied)	611239	2476	6296	620011
VAL Demo Application da_val_*	Kernel-space	3604	528	0	4132
	User-space (combined)	110008	1128	2592	113728
	User-space (proxied)	22910	764	560	24234
Demo Application da_val_specials_*	User-space (combined)	133248	1152	3640	138040
	User-space (proxied)	46683	832	1608	49123
Demo Application da_securedebug_*	User-space (combined)	124640	1152	2736	128528
	User-space (proxied)	37542	788	704	39034
Demo Application da_otpprogram_*	User-space (combined)	101913	1128	4712	107753
	User-space (proxied)	15116	796	2680	18592
Demo Application da_otpkeyblob_*	User-space (combined)	121309	1128	9288	131725
	User-space (proxied)	34512	796	7256	42564

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